Computer Science

(A Level, OCR)

Computer Science is a practical subject where students can apply the academic principles learned in the classroom to real world systems. While the assessment is based heavily on two paper based final exams, the course is very much centred on practical programming and students spend the majority of their time developing programming skills. It is an intensely creative subject and one that really develops the students' problem-solving skills by learning about something called computational thinking. This is a tool kit for finding solutions for big problems. A skill that is very transferable.

- The aims of this qualification are to enable students to develop:
- an understanding of, and ability to apply, the fundamental principles and concepts of computer science including; abstraction, decomposition, logic, algorithms and data representation
- the ability to analyse problems in computational terms through practical experience of solving such problems including writing programs to do so
- the capacity for thinking creatively, innovatively, analytically, logically and critically
- the capacity to see relationships between different aspects of computer science
- mathematical skills
- the ability to articulate the individual (moral), social (ethical), legal and cultural opportunities and risks of digital technology

Entry requirements

Grade 6 in GCSE Computer Science
(Grade 6 in GCSE Mathematics if Computer Science not previously studied).

Unit	%	When?	Content
Unit 1 Computer Systems	40%	June of Year 13 2 ½ hour exam 140 marks	Characteristics of contemporary processors. Software and software development. Programming Exchanging data. Data types, structures and algorithms. Legal, moral, ethical and cultural issues.
Unit 2 Algorithms and programming	40%	June of Year 13 2 ½ hour exam 140 marks	Elements of computational thinking. Programming and problem solving. Pattern recognition, abstraction and decomposition. Algorithm design and efficiency. Standard algorithms.
Unit 3 Programming Project	20%	Coursework submitted in March of Year 13 70 marks	Analysis of a problem to enable students to demonstrate the skills and knowledge necessary to meet the assessment objectives. Students will need to analyse the problem, design a solution, implement the solution and give a thorough evaluation.

Career Links

This course is ideal for students wishing to pursue Computer Science, Information Systems, Multimedia, Software Engineering, Computer Networking, e-Business and Information Management at degree level, or for anyone considering any kind of career in computing. It is also a good additional subject for any student considering taking Mathematics, Engineering or Sciences.

For more information please contact Mrs M Hepworth.